

assignment _torques basics

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:44pm

Quiz Instructions



Question 1 14.5 pts

Two forces produce equal torques on a door about the door hinge. The first force is applied at the midpoint of the door; the second force is applied at the doorknob. Both forces are applied perpendicular to the door. Which force has a greater magnitude?

hint: Same torque but different force and distance. So if distance goes down, force has to go up because torque = force x distance.



the first force (at the midpoint)



the second force (at the doorknob)



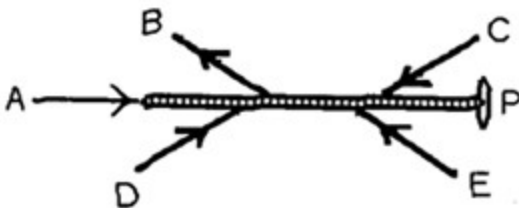
The two forces are equal.



Question 2 14.5 pts

Five forces act on a rod that is free to pivot at point P, as shown in the figure. Which of these forces is producing a counter-clockwise torque about point P? (There could be more than one correct choice.)

hint: it helps if you draw the perpendicular component to see in which direction it will rotate.



force C



force D



force B



force A



force E



Question 3 14.5 pts

If the sum of the external torques on an object is zero, then the sum of the external forces on it must also be zero.

hint: go back to my slides. What is the condition for equilibrium.



False



True



Question 4 14.5 pts

The lug nuts on a car wheel require tightening to a torque of $90 \text{ N} \cdot \text{m}$. If a 30-cm long wrench is used, what is the magnitude of the *minimum* force required using the wrench?

hint: you get the minimum force when its perpendicular to the lever.



300 N



150 N



30 N



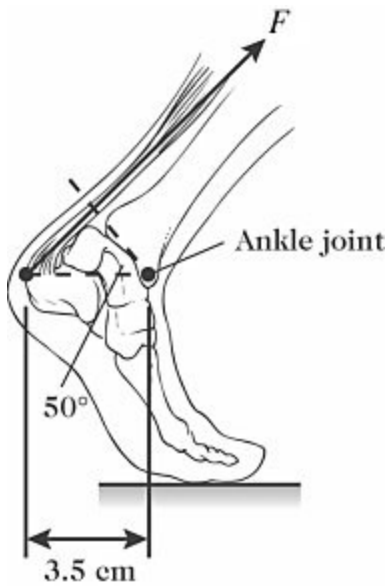
15 N



Question 5 14.5 pts

The figure shows a person's foot. In that figure, the Achilles tendon exerts a force of magnitude $F = 720 \text{ N}$. What is the magnitude of the torque that this force produces about the ankle joint?

hint: refer to my slides



☐ 12 N · m

☐ 16 N · m

☐ 36 N · m

☐ 21 N · m

☐ 26 N · m



Question 6 14.5 pts

A man in a gym is holding an 8.0-kg weight at arm's length, a distance of 0.55 m from his shoulder joint. What is the torque about his shoulder joint due to the weight if his arm is held at 30° below the horizontal?

hint: draw a horizontal line. Then draw a line 30 degrees below (clockwise). That's your arm and the corner is your shoulder about which the arm can rotate. Draw a mass at the end of your arm. At the mass draw the weight straight down (vertical. Its an arrow). . Draw the components of the weight from the mass (tail of the weight). Can you see that the perpendicular component and the weight make a 30 degrees angle with the weight (vertical). Find the torque.

☐ 2.2 N · m

☐ 13 N · m

☐ 22 N · m

☐ 37 N · m



4.4 N · m



Question 7 14.5 pts

Two equal-magnitude forces are applied to a door at the doorknob. The first force is applied perpendicular to the door, and the second force is applied at 30° to the plane of the door. Which force exerts the greater torque about the door hinge?



Both forces exert equal non-zero torques.



the second force (applied at an angle)



Both forces exert zero torque.



the first force (applied perpendicular to the door)

Not saved

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